

Arjun Sidhu

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CS graduate with 4 co-op terms building AI-enabled software, RAG pipelines, agent evaluation systems, and workflow automation platforms

PROFESSIONAL EXPERIENCE

Software Engineer (Co-op)

Jan 2025 - Aug 2025

Ministry of Public and Business Service Delivery and Procurement (MPBSDP)

Toronto, ON

- Built internal workflow platforms used by 15+ stakeholders, reducing manual processing time by ~45% across reporting, CRM, approvals, and call tracking workflows
- Translated ambiguous business workflows into production requirements, data models, and automated systems using Power Apps, Power Automate, and SharePoint

Software Engineer (Co-op)

May 2024 - Aug 2024

Ontario Ministry of Transportation (MTO)

Toronto, ON

- Designed and shipped internal automation tools for approval routing, data collection, and workflow management using Power Platform
- Reduced manual processing time by ~50% by replacing repetitive business processes with automated forms, routing logic, and reporting workflows

AI/ML Developer (Co-op)

Sept 2023 - Dec 2023

Ontario Ministry of Transportation (MTO)

Toronto, ON

- Built an Azure OpenAI text-to-SQL system enabling natural language querying of PostgreSQL databases, handling prompt construction, query parsing, and result formatting
- Developed a Streamlit AI assistant for automated data retrieval, analysis, and structured reporting from internal datasets
- Improved response accuracy and reduced API cost by ~20% through prompt engineering and optimization

PROJECTS

AgentEval Harness | FastAPI, React, TypeScript, SQLite/Postgres, Python

2026

- Built a full-stack agent evaluation platform for 20+ RAG and tool-using workflows, measuring correctness, groundedness, hallucination signals, tool-use accuracy, latency, and cost
- Implemented modular scoring engine, retrieval inspection, action validation, pytest coverage, and React dashboard to compare runs and diagnose retrieval, reasoning, and tool-selection failures

ChangeFlow Platform | React, TypeScript, Node.js, PostgreSQL, AWS S3, Claude API, Voyage AI

- Built an agentic document platform with a custom LLM orchestration layer that uses Claude API to analyze documents, extract risks, and route decisions to structured action handlers for task creation, alerts, and approval workflows
- Designed RAG pipeline with Voyage AI embeddings, PostgreSQL storage, cosine similarity retrieval, keyword fallback, and multi-step LLM orchestration

Fill Shift | Next.js, TypeScript, Supabase, Postgres, Twilio, Vitest, Vercel

2026

- Built a staffing-risk platform that detects 48–72 hour shift gaps, recommends eligible staff, and supports human-approved SMS outreach
- Designed multi-tenant Supabase/Postgres schema with auth, row-level security, scheduled gap detection, audit trails, CSV imports, and test coverage

EDUCATION

Bachelor of Computing Honours, Computer Science (Co-op)

Sept. 2021 – Apr. 2026

University of Guelph

Guelph, ON

- Relevant Courses: Discrete Structures 1 & 2, Computer Networks, Software Engineering, Parallel Programming, Operating Systems, Computational Intelligence, Theory of Computation, Compilers, Computer Graphics

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++

AI/ML: RAG, LLM Orchestration, Agent Evaluation, Prompt Engineering, Agentic Workflows, Embeddings

Frameworks & Libraries: React, Next.js, Vite, Node.js, Express, FastAPI, REST APIs, PyTorch, Scikit-Learn

Databases & Cloud: PostgreSQL, SQLite, Supabase, AWS S3, Azure, pgvector

Tools & Platforms: Git, Docker, Pytest, Vitest, Azure OpenAI, Claude API, OpenAI API, Voyage AI, Vercel